

Objects and Data

Deliverables from this lab:

1. A database class that can be used by other object classes for interacting with the database

What are you doing?

Your final project must connect to a database of some kind. This lab will require the creation of a database class you can use with any other object class to interact with the database, including all Create/Read/Update/Delete (CRUD) operations.

The purpose of the database class is to provide a consistent way for interacting with the database throughout the project, and to enforce code reuse instead of a bunch of object classes and/or inline code duplicating functionality; if/when the functionality needs to change (e.g. database connection details, etc.), it only needs to be updated in one place instead of many throughout the project code files.

1. Create a database class for other object classes to use for interacting with the database, including at least the following methods:
 - a. Connect()
 - i. Establishes a connection to the database using the required login credentials (e.g. database server, database name, username, password)
 - b. Disconnect()
 - i. Disconnects from the database
 - c. Execute()
 - i. Executes an SQL statement passed into the method as an argument
 - ii. NOTE: Use parameterized queries to minimize SQL injection
 - iii. HINT: Pass an optional argument into the method using an associative array or list to bind parameters
2. Create the following methods in at least one object class to use the database class:
 - a. Save()
 - i. Inserts a new record into the database ----OR----
 - ii. Updates an existing record if passed an "id"
 - b. Load()
 - i. Reads a record from the database using an "id" primary key
 - c. Delete()

- i. Deletes a record from the database using an “id” primary key

Getting Done Early

Take advantage of the class time to utilize your database class in more object classes. You can use the database to store user preferences and various configuration settings.

SHARING YOUR WORK

1. Share your database class with your table
 - a. Choose the table consensus as the “best” database class at your table to receive a truffle
2. A table representative will be randomly chosen to show their database class with the entire class
 - a. NOTE: the table representative may or may not be the same person chosen with the best database class at the table